

Krishna Jalan

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Research Interests

Semi-classical gravity; Black hole thermodynamics; Holography; Quantum information aspects of quantum field theory and gravity.

Education

- **Institute of Mathematical Sciences** Chennai, India
Ph.D. in Theoretical Physics Sep 2020 - present
 - Thesis: Black Hole Dynamics & Holography
 - Advisor: Dr. Roji Pius
- **National Institute of Technology** Rourkela, India
(Integrated) M.Sc. in Physics, CGPA = 9.02/10 Aug 2014 – May 2019
 - Thesis: Anisotropic Thermalization in Strongly Coupled Field Theory via Gauge/Gravity Duality
 - Advisor: Dr. Subhash Chandra Mahapatra

Work Experience

- **Gauhati University** Guwahati, India
Research Assistant Sep 2019 – Dec 2019
Project on early universe dynamics related to inflation theory and origin of density perturbations, and dynamical dark energy under the guidance of *Dr. Sanjeev Kalita*.
- **National Institute of Technology** Rourkela, India
Master's Student Aug 2018 – May 2019
Master's thesis project on the application of gauge/gravity duality to study anisotropic signature of magnetic field on the thermalization time of quark-gluon plasma.
- **Harish-Chandra Research Institute** Prayagraj, India
Summer Research Fellow May 2018 – Jul 2018
Project on dark matter particles titled *A Study of Dark Matter* under the guidance of *Prof. Biswarup Mukhopadhyaya*.
- **Jawaharlal Nehru University** Delhi, India
Summer Research Fellow May 2017 – Jul 2017
Project on neutrino oscillations titled *Phenomenology of Neutrino Oscillation* under the guidance of *Dr. Poonam Mehta*.

Publications

Peer-reviewed

2. Krishna Jalan, Roji Pius, and Manish Ramchander. “Island Paradigm and Information Recovery from Radiation,” *JHEP* **07** (2025) 244. [2404.00773].
1. Prateksh Dhivakar and Krishna Jalan. “Generalized second law for non-minimally coupled matter theories,” *Class. Quant. Grav.* **41** (2024) 15, 155003. [2309.12782]

Under Review/Preprint

1. Krishna Jalan and Roji Pius. “Reduced Half-sided Translations and Islands,” 2023. [\[2312.11085\]](#).

Presentations

Invited Talks

- **An Introduction to Black Hole Thermodynamics**

- *Student Talks on Trending Topics in Theory*, Indian Institute of Science Education and Research, Bhopal, India *July 2025*
Gave a set of 6 lectures on several aspects of black hole thermodynamics.

- **Algebraic approach to QFT**

- *Amplitudes@IITI 25*, Indian Institute of Technology, Indore, India *January 2025*
Gave a set of 4 talks on algebraic quantum field theory and type classification of von Neumann algebras.

- **Generalized Second Law for Non-minimally Coupled Matter Theories**

- *Chennai Strings Meeting*, jointly organized by CMI-IMSc-IIT Madras, India *November 2024*

- **Half-sided Translations and Islands**

- *Chennai Strings Meeting*, jointly organized by CMI-IMSc-IIT Madras, Chennai, India *November 2023*

Poster Presentation

- **Island paradigm and information recovery from radiation**

- *Indian Strings Meeting 2025*, NISER-IIT Bhubaneswar, India *December 2025*

- **Generalized Second Law for Non-minimally Coupled Matter Theories**

- *National Strings Meeting 2024*, IIT Ropar, India *December 2024*
- *Quantum Extreme Universe: Matter, Information, and Gravity Workshop*, Okinawa Institute of Science & Technology, Okinawa, Japan *October 2024*
- *Quantum Information, Quantum Field Theory, and Gravity Workshop*, International Centre for Theoretical Sciences, Bengaluru, India *August 2024*
- *3rd Quantum Field Theory in Curved Spacetimes Workshop*, Instituto Superior Técnico, Lisbon, Portugal *June 2024*
(Awarded a sum of 1500 Euros, courtesy of the [Julian Schwinger Foundation for Physics Research](#), to support my participation)

- **Half-sided Translations and Islands**

- *Indian Strings Meeting 2023*, IIT Bombay, Mumbai, India *December 2023*

Miscellaneous

- **Generalized Second Law for Non-minimally Coupled Matter Theories**

- *Department Seminar*, NIT Rourkela, India *December 2025*
- *Entanglement Meeting*, YITP, Kyoto, Japan *November 2024*

- *Qubits and Spacetime Unit Seminar*, OIST, Okinawa, Japan October 2024
- *Department Seminar*, IIT Kanpur, India September 2024
- **Introduction to Covariant Phase Space: the Geometry of “Geometry”**
 - Chennai Mathematical Institute, Chennai, India February 2024
- **Black Hole Interiors and Modular Inclusions**
 - Institute of Mathematical Sciences, India May 2023
- **Algebras and Emergent “Times”**
 - Institute of Mathematical Sciences, India May 2022

Selected Conferences/Workshops

Excluding the ones where I have given/presented a talk/poster.

- *Quantum Gravity and Information Theory: Modern Developments*, Chennai Mathematical Institute, India, November 2025.
- *1st Advanced Lectures in Physics in Switzerland*, Les Diablerets, Switzerland, May 2024. Partially funded by the sponsors – **SwissMAP** and **Quantinuum**.
- *Young Researchers’ Gravity Meeting*, Chennai Mathematical Institute, India, September 2023.
- *Observable Algebras in Field Theory and Gravity, JulyPhy Workshop*, Online Conference, IIT Mandi, India, July 2023.
- *Student Talks on Trending Topics in Theory (ST4 2023)* Workshop, IIT Mandi, India, July 2023.
- *Black Holes and Gauge Theories with Holographic Enlightenment*, IIT Madras, Chennai, India, January 2023.
- *Non-perturbative and Numerical Approaches to Quantum Gravity, String Theory, and Holography*, ICTS-TIFR, Bengaluru, India, August-September 2022.
- *Quantum Field Theory in Curved Spacetimes Workshop*, Online Conference, Spain, May 2022.
- *Indian Strings Meeting 2021*, Online Conference, IIT Roorkee, India, December 2021.
- *All Lambdas Holography @ Prague*, Online Conference, Czechia, October 2021.
- *Black Holes Inside and Out*, Online Conference, Japan, September 2021.

Awards & Achievements

- Qualified for the prestigious [INSPIRE Fellowship](#) awarded by Department of Science and Technology, Government of India, for the year 2020, for carrying out doctoral degree in both basic and applied sciences. (did not avail)
- Qualified the following national level entrance tests for *Junior Research Fellowship* eligibility – UGC-CSIR National Eligibility Test, Graduate Aptitude Test in Engineering (Physics), and Joint Entrance Screening Test, 2020.
- Institute Gold Medal for the Best Postgraduate with Integrated M.Sc. Degree among the natural sciences, NIT Rourkela, awarded in January 2020.

- Institute Silver Medal for the Best Postgraduate with Integrated M.Sc. Degree in Department of Physics & Astronomy, NIT Rourkela awarded in January 2020.
- Selected for the prestigious *Summer Research Fellowship*, awarded jointly by the academies of sciences in India (IASc-INSANA-SASI), for two consecutive years, May - July in 2017 and 2018.
- Recipient of Academic Excellency Award, Department of Physics and Astronomy, NIT Rourkela, for 4 consecutive academic years, 2014-15, 2015-16, 2016-17 and 2017-18.
- [INSPIRE Scholarship](#) awarded by Department of Science and Technology, Government of India, to the top 1 percentile students of the country based on Higher Secondary merit, February 2015 - May 2019.
- Qualified *Regional Mathematics Olympiad* from the North Eastern Region of India, 2013.
- Awarded the "*Certificate of Appreciation*" for academic excellence in the Matriculation examination, by the Deputy Commissioner, Chirang District, Assam, India, 2012.
- Gold Medal in Mathematics Olympiad conducted by All India Schools Mathematics Teachers Association, 2011.

Teaching Experience*

* All the courses assisted at the Institute of Mathematical Sciences (IMSc) were graduate level courses. The duties involved preparation and grading of assignments and conducting tutorial sessions for a class size of about 10-20 students.

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| • Quantum Field Theory I
<i>Teaching Assistant</i> | IMSc
<i>Fall 2025</i> |
| • Quantum Mechanics II
<i>Teaching Assistant</i> | IMSc
<i>Spring 2024</i> |
| • Electromagnetic Theory
<i>Teaching Assistant</i> | IMSc
<i>Fall 2023</i> |
| • Quantum Field Theory II
<i>Teaching Assistant</i> | IMSc
<i>Spring 2023</i> |
| • Quantum Mechanics I
<i>Teaching Assistant</i> | IMSc
<i>Fall 2022</i> |
| • General Relativity
<i>Teaching Assistant</i> | IMSc
<i>Spring 2022</i> |
| • Indian Olympiad Qualifier in Mathematics training camp
Taught an online problem solving session organized by the <i>Assam Academy of Mathematics, Bongaigaon Branch</i> . | <i>October 2022</i> |
| • Freshmen Physics Remedial Class
Tutor under Student Counselling Program at NIT Rourkela. | <i>Aug 2017 - Apr 2018</i> |

Skills

- **Technical:**
 - OS → Linux, Windows.

- **Coding** → C (basic), C++ (basic), FORTRAN (intermediate), Python (intermediate), Mathematica (intermediate), Maxima (basic).
- **Documentation** → L^AT_EX (advanced), Office (advanced).
- **Languages:**
Marwadi (mother tongue), Hindi (fluent), English (fluent), Assamese (basic), Bengali (basic), Japanese (basic).

Non-academic Work

- Part of IMSc outreach team, where I have helped organize several astronomy related events for general public, particularly school students. *Sep 2024 – present*
- **Mentor, Student Counselling Program** *Aug 2017 – Apr 2018*
Guiding a group of four first-year undergraduate students on academic and non-academic matters during their freshmen year.